

Correct Answers, Uneven Grounding

The distributed-evidence gap in legal RAG

Answers remain similarly accurate. Complete source support falls when evidence spans pages.

-12.2 pp
page-grounding gap
95% CI -19.0 to -5.9

95 eligible questions within a 100-question benchmark / 30 PDFs / 590 pages

1 HYPOTHESIS

A model can answer from retrieved evidence yet emit an incomplete source set. Aggregate scores can hide whether this mismatch depends on evidence shape. [1,4,5]

RESEARCH QUESTION
How does distributed evidence affect the relationship between answer quality and source grounding in a legal RAG pipeline?

H1
Multi-page evidence will preserve answer quality but reduce final page grounding. Uniform citation narrowing may widen that gap.

59 localized exactly one gold page 36 distributed multiple gold pages

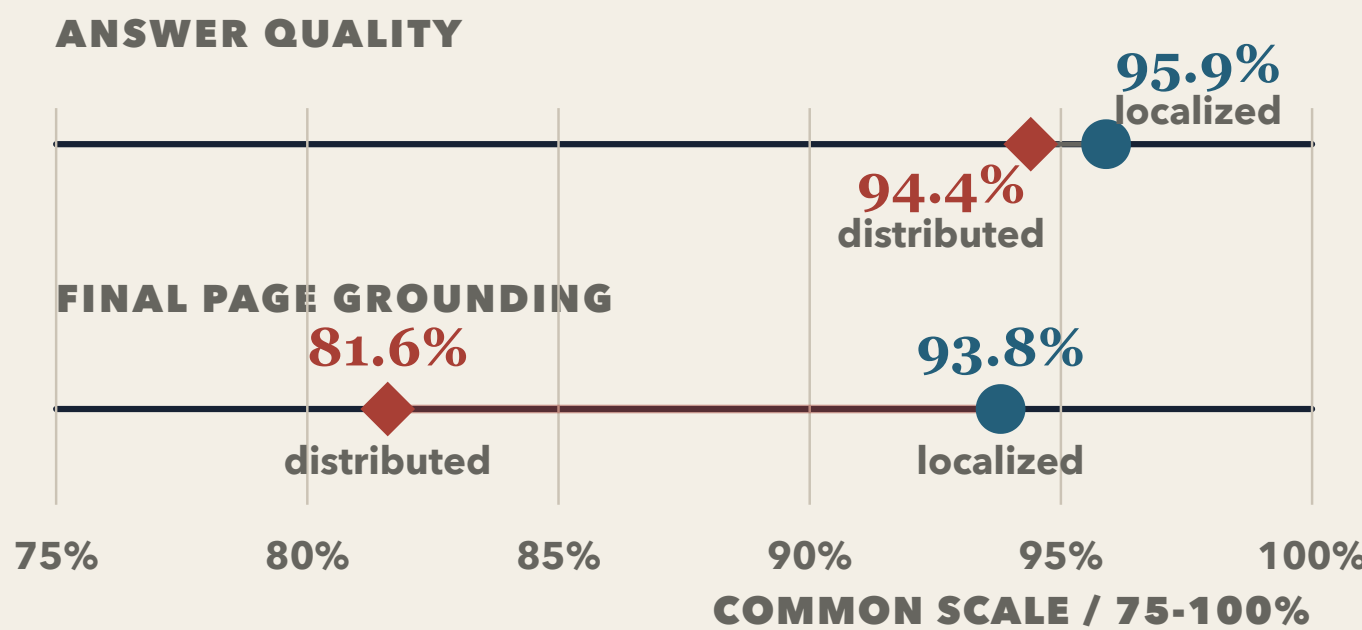
Answer quality / mean benchmark score on a 0-1 scale.
Source grounding / page F-beta, beta = 2.5, weighting recall.
Evidence shape / 35 of 36 distributed questions also span documents; the effects cannot be separated.

2 METHODOLOGY

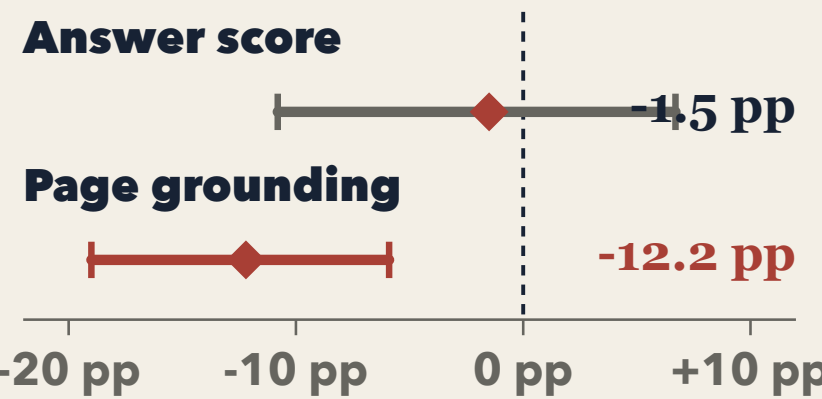


3 RESULTS

Answers stay level.
Grounding separates.



DISTRIBUTED MINUS LOCALIZED



20,000 stratified bootstrap resamples. The answer interval crosses zero; the grounding interval does not.

PERFECT ANSWER / GROUNDING BELOW 0.8

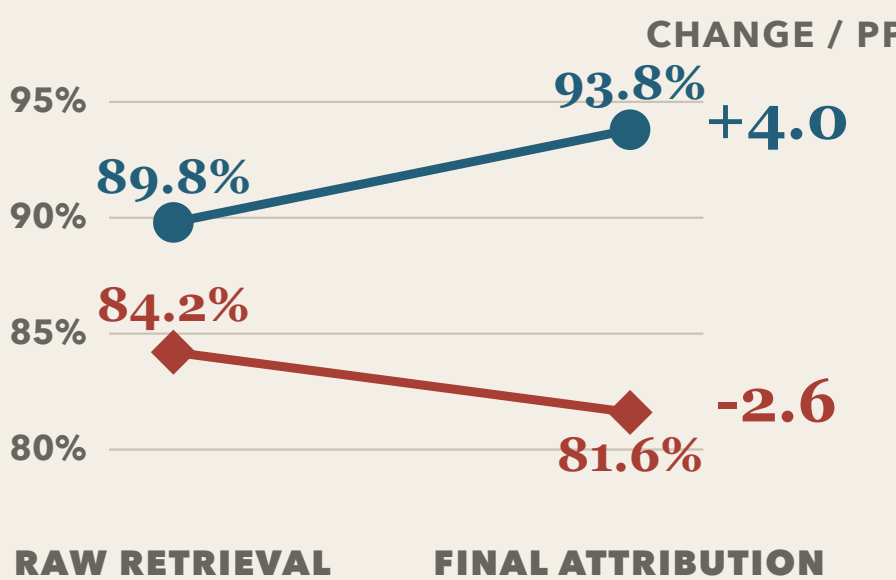
Localized 4 / 55 7.3%
Distributed 16 / 34 47.1%

Among answers scoring 1.0, incomplete page support is much more common for distributed evidence.

4 RESULTS / DIAGNOSTIC

Citation narrowing moves groups apart.

Page-level F-beta from the same run, before and after answer-conditioned attribution.



DECISION

Use evidence spread as a routing signal: narrow localized support; preserve and aggregate distributed support.

Gold evidence shape is unavailable at inference time. Use observable proxies: retrieval spread, cross-document dependencies, or claim-level support checks.

LIMITATIONS

One saved development run; retrospective hypothesis; observational groups; page and document spread are confounded; answer types and scoring methods are uneven; the grounding-below-0.8 threshold is descriptive; stage changes are not a causal ablation.

REPRODUCIBILITY

github.com/SteinVR/Agentic-RAG-Challenge
20,000 stratified bootstrap resamples / seed 20260903. Saved-run artifacts include `solver_narrowing_true.yaml`, `benchmark_answer_scores.checkpoint.jsonl`, and `grounding/ledger.csv`.

SELECTED REFERENCES

[1] Lewis et al., NeurIPS 2020 / RAG. [2] Karpukhin et al., EMNLP 2020 / dense retrieval. [3] Cormack et al., SIGIR 2009 / RRF. [4] Gao et al., EMNLP 2023 / citations. [5] Pipitone and Houir Alami, 2024 / LegalBench-RAG. [6] Zhang et al., 2025 / Qwen3 Embedding.